

Moving Charges & Magnetism: A Visual Revision

FUNDAMENTAL CONCEPTS

Electric Current Creates a Magnetic Field
 Serenated discovery: current-carrying conductor is surrounded by a magnetic field. (Magnetic effect of current)

Right-Hand Thumb Rule

Stationary Charge: Electric Field Only
 Moving Charge: Electric & Magnetic Fields
 Current Conductor: Magnetic Field Only (No Net Charge)

FORCE ON A MOVING CHARGE (LORENTZ FORCE)

Force is Zero:
 $v=0$ $\theta=0^\circ$ $\theta=180^\circ$
 $F=0$ $F=0$ $F=0$

Maximum Force Condition:
 $\theta=90^\circ$
 $F = qvB$

Fleming's Left-Hand Rule

Force on a Moving Charge (Lorentz Force)
 $F = qvB\sin\theta$
 (or $F = q(\mathbf{v} \times \mathbf{B})$)

MOTION OF A CHARGE IN A MAGNETIC FIELD

Path 1: Perpendicular Entry ($\theta=90^\circ$)
 Follows a circular path. Speed & KE constant, velocity changes.
 Radius (r) = $\frac{mv}{qB}$
 Time Period (T) = $\frac{2\pi m}{qB}$

Path 2: Entry at an Angle (θ)
 Helical path. $v\sin\theta$ causes circular motion, $v\cos\theta$ causes linear motion.
 Radius (r) = $\frac{mv\sin\theta}{qB}$
 Time Period (T) = $\frac{2\pi m}{qB}$ (Independent of velocity!)
 Pitch (P) = $(v\cos\theta) \cdot T$

CALCULATING MAGNETIC FIELDS: KEY LAWS

Biot-Savart Law

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{l} \times \mathbf{r}}{r^3}$$

Ampere's Circuital Law

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enclosed}}$$

APPLICATIONS & DEVICES

Field at the Center of a Coil

$$B = \frac{\mu_0 N I}{2r}$$

Force Between Parallel Wires

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

Solenoid

$$B = \mu_0 n I, \quad n = \frac{\text{turns}}{\text{length}}$$

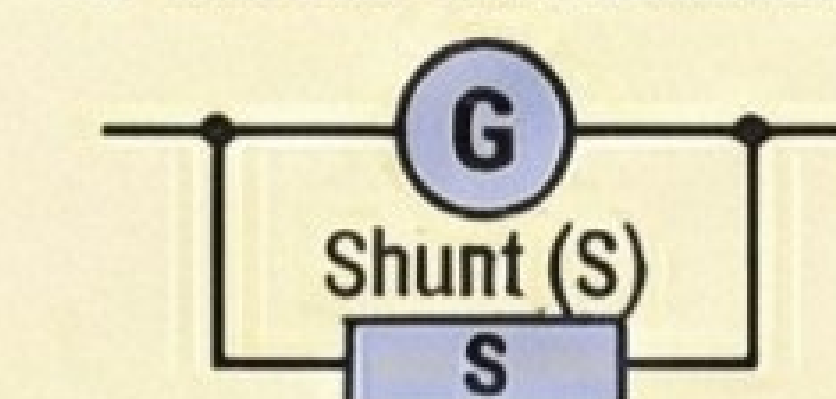
Torque on a Current Loop

$$\tau = N I A B \sin\theta$$

 ($M = N I A$, magnetic moment)

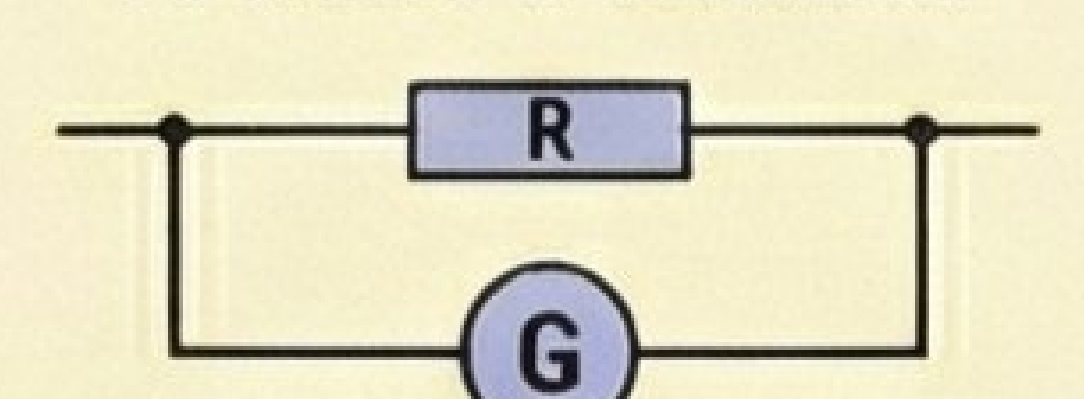
CONVERTING A GALVANOMETER

To create an Ammeter



Connect low resistance (S) in parallel to divert most current.

To create a Voltmeter



Connect high resistance (R) in series to limit current flow.